



Titanium Grade 2

Categories: [Metal](#); [Nonferrous Metal](#); [Titanium Alloy](#); [Unalloyed/Modified Titanium](#)

Material Information provided by Allvac and the references.

Notes: **Characteristics:** Higher ductility, lower strength when compared to Ti alloys

Applications: Airframe components, cryogenic vessels, heat exchangers, CPI equipment, condenser tubing, pickling baskets.

Forms typically available: billet, bar, plate, sheet, strip, rod, wire, extrusions, tubing.

Titanium Grades 1,2,3,4,7,11,and 12 are all considered unalloyed and have similar mechanical properties.

Key Words: ASTM Grade 2; UNS R50400, CP titanium, C.P. titanium alloy

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Physical Properties	Metric	English	Comments
Density	4.51 g/cc	0.163 lb/in ³	
Chemical Properties	Metric	English	Comments
Thermal Neutron Cross Section	5.6 barns/atom	5.6 barns/atom	Average for naturally occurring Ti
Mechanical Properties	Metric	English	Comments
Hardness, Brinell	200	200	
Hardness, Knoop	170	170	
Hardness, Rockwell B	80	80	
Hardness, Vickers	145	145	
Tensile Strength 	103 - 138 MPa @Temperature 540 °C	14900 - 20000 psi @Temperature 1000 °F	
	131 - 186 MPa @Temperature 425 °C	19000 - 27000 psi @Temperature 797 °F	
	172 - 207 MPa @Temperature 315 °C	24900 - 30000 psi @Temperature 599 °F	

	207 - 228 MPa @Temperature 205 °C	30000 - 33100 psi @Temperature 401 °F	
Tensile Strength, Ultimate	344 MPa	49900 psi	
Tensile Strength, Yield	276 - 448 MPa	40000 - 65000 psi	
	62.0 - 76.0 MPa @Temperature 540 °C	8990 - 11000 psi @Temperature 1000 °F	0.2% offset
	76.0 - 103 MPa @Temperature 425 °C	11000 - 14900 psi @Temperature 797 °F	0.2% offset
	90.0 - 124 MPa @Temperature 315 °C	13100 - 18000 psi @Temperature 599 °F	0.2% offset
	138 - 152 MPa @Temperature 205 °C	20000 - 22000 psi @Temperature 401 °F	0.2% offset
Elongation at Break	20 %	20 %	
	24 - 28 % @Temperature 425 °C	24 - 28 % @Temperature 797 °F	In 50 mm
	30 - 35 % @Temperature 540 °C	30 - 35 % @Temperature 1000 °F	In 50 mm
	30 - 38 % @Temperature 315 °C	30 - 38 % @Temperature 599 °F	In 50 mm
	38 - 45 % @Temperature 205 °C	38 - 45 % @Temperature 401 °F	In 50 mm
Reduction of Area	30 %	30 %	
	55 - 65 % @Temperature 205 °C	55 - 65 % @Temperature 401 °F	
	70 - 80 % @Temperature 315 °C	70 - 80 % @Temperature 599 °F	
	70 - 80 % @Temperature 425 °C	70 - 80 % @Temperature 797 °F	
	70 - 80 % @Temperature 540 °C	70 - 80 % @Temperature 1000 °F	
Tensile Modulus	103 GPa	14900 ksi	
Compressive Modulus	110 GPa	16000 ksi	
Poissons Ratio	0.37	0.37	
Fatigue Strength 	300 MPa @# of Cycles 1.00e+7	43500 psi @# of Cycles 1.00e+7	unnotched
	425 MPa @# of Cycles 30000	61600 psi @# of Cycles 30000	Unnotched
Fracture Toughness	66.0 MPa-m ^{1/2}	60.1 ksi-in ^{1/2}	K(Q); annealed
Shear Modulus	45.0 GPa	6530 ksi	
Izod Impact	114 - 171 J	84.1 - 126 ft-lb	
Charpy Impact 	100 J @Temperature 200 °C	73.8 ft-lb @Temperature 392 °F	Notched
	115 J @Temperature 300 °C	84.8 ft-lb @Temperature 572 °F	Notched
	115 J @Temperature 100 °C	84.8 ft-lb @Temperature 212 °F	Notched
	160 J @Temperature 0.000 °C	118 ft-lb @Temperature 32.0 °F	Notched
	175 J @Temperature -100 °C	129 ft-lb @Temperature -148 °F	Notched
Coefficient of Friction, Dynamic	0.68	0.68	Ti sliding on Ti; 300 m/min
	0.80	0.80	Ti sliding on Ti; 40 m/min
Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000520 ohm-cm	0.0000520 ohm-cm	
Magnetic Permeability	1.00005	1.00005	at 1.6 kA/m
Magnetic Susceptibility	4.0e-8	4.0e-8	kg; Paramagnetic

Thermal Properties	Metric	English	Comments
Heat of Fusion	325 J/g	140 BTU/lb	High Purity Ti.
CTE, linear 	8.60 $\mu\text{m/m-}^{\circ}\text{C}$ @Temperature 0.000 - 100 $^{\circ}\text{C}$	4.78 $\mu\text{in/in-}^{\circ}\text{F}$ @Temperature 32.0 - 212 $^{\circ}\text{F}$	
	9.20 $\mu\text{m/m-}^{\circ}\text{C}$ @Temperature 0.000 - 315 $^{\circ}\text{C}$	5.11 $\mu\text{in/in-}^{\circ}\text{F}$ @Temperature 32.0 - 599 $^{\circ}\text{F}$	average
	9.70 $\mu\text{m/m-}^{\circ}\text{C}$ @Temperature 0.000 - 540 $^{\circ}\text{C}$	5.39 $\mu\text{in/in-}^{\circ}\text{F}$ @Temperature 32.0 - 1000 $^{\circ}\text{F}$	perpendicular to the c-axis
Specific Heat Capacity 	0.523 J/g- $^{\circ}\text{C}$ @Temperature 20.0 $^{\circ}\text{C}$	0.125 BTU/lb- $^{\circ}\text{F}$ @Temperature 68.0 $^{\circ}\text{F}$	
	0.560 J/g- $^{\circ}\text{C}$ @Temperature 200 $^{\circ}\text{C}$	0.134 BTU/lb- $^{\circ}\text{F}$ @Temperature 392 $^{\circ}\text{F}$	
	0.620 J/g- $^{\circ}\text{C}$ @Temperature 400 $^{\circ}\text{C}$	0.148 BTU/lb- $^{\circ}\text{F}$ @Temperature 752 $^{\circ}\text{F}$	
	0.670 J/g- $^{\circ}\text{C}$ @Temperature 540 $^{\circ}\text{C}$	0.160 BTU/lb- $^{\circ}\text{F}$ @Temperature 1000 $^{\circ}\text{F}$	
	0.690 J/g- $^{\circ}\text{C}$ @Temperature 600 $^{\circ}\text{C}$	0.165 BTU/lb- $^{\circ}\text{F}$ @Temperature 1110 $^{\circ}\text{F}$	
Thermal Conductivity	16.4 W/m-K	114 BTU-in/hr-ft 2 - $^{\circ}\text{F}$	annealed
Melting Point	≤ 1665 $^{\circ}\text{C}$	≤ 3029 $^{\circ}\text{F}$	
Liquidus	1665 $^{\circ}\text{C}$	3029 $^{\circ}\text{F}$	
Beta Transus	913 $^{\circ}\text{C}$	1680 $^{\circ}\text{F}$	

Optical Properties	Metric	English	Comments
Emissivity (0-1)	0.30 @Temperature 710 $^{\circ}\text{C}$	0.30 @Temperature 1310 $^{\circ}\text{F}$	High purity Ti
Reflection Coefficient, Visible (0-1)	0.56	0.56	High purity Ti; visible light.

Component Elements Properties	Metric	English	Comments
Carbon, C	≤ 0.10 %	≤ 0.10 %	
Hydrogen, H	≤ 0.015 %	≤ 0.015 %	
Iron, Fe	≤ 0.30 %	≤ 0.30 %	
Nitrogen, N	≤ 0.030 %	≤ 0.030 %	
Other, total	≤ 0.30 %	≤ 0.30 %	
Oxygen, O	≤ 0.25 %	≤ 0.25 %	
Titanium, Ti	≥ 98.885 %	≥ 98.885 %	

Descriptive Properties

Crystal Structure	bcc cph	above 913 $^{\circ}\text{C}$ below 890 $^{\circ}\text{C}$
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[References](#) for this datasheet.

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